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Studierichting: Informatica
Oefeningenreeks 4A nr. 22.4

$$\int x\sqrt{1-x^2}\,\mathrm{Bgsin}(x)dx$$

$$\begin{cases} u = \mathrm{Bgsin}(x) \\ dv = x\sqrt{1-x^2}dx \end{cases} \Rightarrow \begin{cases} du = \dfrac{1}{\sqrt{1-x^2}}dx \\ v = \dfrac{-(1-x^2)^{\frac{3}{2}}}{3} \end{cases}$$

$$= \frac{-(1-x^2)^{\frac{3}{2}}\,\mathrm{Bgsin}(x)}{3} - \int \frac{-(1-x^2)dx}{3}$$

$$= \frac{-(1-x^2)^{\frac{3}{2}}\,\mathrm{Bgsin}(x)}{3} - \left(\frac{1}{3}\int x^2dx - \frac{1}{3}\int 1dx\right)$$

$$= \frac{-(1-x^2)^{\frac{3}{2}}\,\mathrm{Bgsin}(x)}{3} - \frac{x^3}{9} + \frac{x}{3} + \mathrm{C}$$

References